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Semester – V

Business Analytics

Experiment No.: 4

Cleaning and Preparing Business Data Using Power Query

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Cleaning and Preparing Business Data Using Power Query

Problem Statement

A company has received customer data containing missing values, duplicate records, and inconsistent entries. Clean and prepare the dataset using Power Query before analysis.

Methodology / Steps Followed

Step 1: Data Collection

- Obtained the **Dirty Cafe Sales dataset** in CSV format.
- The dataset contains **10,000 transaction records**.
- The dataset contains the following fields:
 - Transaction ID
 - Item
 - Quantity
 - Price Per Unit
 - Total Spent
 - Payment Method
 - Location
 - Transaction Date
- The dataset contains customer transaction information that can be prepared for further business analysis.

Step 2: Data Import

Importing the CSV file:

1. Open Power BI Desktop.
2. Select Home → Get Data.
3. Select Text/CSV.
4. Select the dirty_cafe_sales(1).csv file.
5. Preview the dataset.
6. Select Transform Data.

Step 3: Data Cleaning and Transformation

The imported dataset is checked and prepared using Power Query.

1. Check the column names:

Column
Transaction ID
Item
Quantity
Price Per Unit
Total Spent
Payment Method
Location
Transaction Date

2. Check for duplicate records

In Power Query:

1. Select the entire dataset.
2. Go to Home → Remove Rows → Remove Duplicates.

3. Handle missing values

1. Select the column.
2. Go to Transform → Replace Values.
3. Replace null/blank values with Unknown.

4. Handle ERROR and UNKNOWN entries

1. Select the column.
2. Go to Transform → Replace Values.
3. Replace ERROR with Unknown.
4. Repeat for UNKNOWN.

This standardizes the inconsistent entries.

5. Clean Quantity

1. Select Quantity.
2. Replace ERROR with null.
3. Replace UNKNOWN with null.
4. Select the data type icon.
5. Choose Whole Number.

6. Clean Price Per Unit

Replace:

- ERROR → null
- UNKNOWN → null

Then set the data type to:

Decimal Number

7. Clean Total Spent

Replace:

- ERROR → null
- UNKNOWN → null

Then change the data type to:

Decimal Number

8. Clean Payment Method

Standardize these values:

Original	Cleaned
ERROR	Unknown
UNKNOWN	Unknown
Blank	Unknown

The final column should therefore contain consistent categories such as:

Cash | Credit Card | Digital Wallet | Unknown

9. Clean Location

Standardize the inconsistent values:

Original	Cleaned
ERROR	Unknown
UNKNOWN	Unknown
Blank	Unknown

The final categories become:

In-store | Takeaway | Unknown

10. Clean Transaction Date

1. Select Transaction Date.
2. Replace ERROR with null.
3. Replace UNKNOWN with null.
4. Select Data Type → Date.

11. Verify the final data types

Column	Data Type
Transaction ID	Text

Item	Text
Quantity	Whole Number
Price Per Unit	Decimal Number
Total Spent	Decimal Number
Payment Method	Text
Location	Text
Transaction Date	Date

12. Remove unnecessary errors

1. Check each column for remaining ERROR values.
2. Check each column for remaining UNKNOWN values where they shouldn't exist.
3. Check that numerical columns contain only numerical values or nulls.
4. Check that Transaction Date contains valid dates or nulls.
5. Check that categorical fields use consistent naming.

13. Apply the transformations

1. Select Home → Close & Apply.
2. Power BI will load the cleaned dataset into the data model.

Step 4: Data Validation

After loading the cleaned dataset:

1. Open Data View.
2. Check the cleaned columns.
3. Verify that:
 - a. Duplicate records have been checked.
 - b. Missing values have been handled.
 - c. ERROR and UNKNOWN entries have been standardized.
 - d. Numerical columns have appropriate data types.
 - e. Transaction dates are recognized as dates.
 - f. Categorical values are consistent.

Output:

The final Power BI dataset provides:

- Cleaned Transaction Data – Transaction records are prepared for further analysis.
- Missing Value Handling – Missing values are identified and handled appropriately.
- Duplicate Record Check – The dataset is checked for duplicate records.
- Inconsistent Entry Handling – ERROR and UNKNOWN values are standardized.
- Data Type Correction – Numerical and date fields are converted to their appropriate data types.
- Categorical Data Standardization – Fields such as Item, Payment Method, and Location are standardized.

- Analysis-Ready Dataset – The cleaned dataset can now be used to create Power BI visualizations and perform business analysis.

✕ ✓ fx = Table.ReplaceValue(#"Replaced Value26",null,"Unknown",Replacer.ReplaceValue,{"Item"})

Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
Valid 100%	Valid 100%	Valid 100%	Valid 97%	Valid 95%	Valid 95%	Valid 100%	Valid 95%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 3%	Empty 5%	Empty 5%	Empty 0%	Empty 5%

Conclusion

This project successfully cleans and prepares the Dirty Cafe Sales dataset using Power Query in Power BI. The dataset was checked for duplicate records, missing values, inconsistent entries, and incorrect data types.

Missing and inconsistent values such as ERROR and UNKNOWN were handled appropriately, while numerical and date fields were converted into suitable data types. The resulting dataset is standardized and prepared for further analysis and visualization in Power BI.

Learning Outcomes

Through this project, I learned:

- How to import CSV datasets into Power BI.
- How to use Power Query for data cleaning and transformation.
- How to identify missing values in a dataset.
- How to check for duplicate records.
- How to handle inconsistent entries such as ERROR and UNKNOWN.
- How to replace invalid values using Power Query.
- How to change column data types.
- How to clean numerical and date fields.
- How to standardize categorical data.
- How to prepare a dataset for further business analysis.
- How to use Power BI and Power Query to create an analysis-ready dataset.